

2	(a) number of atoms of carbon-12 in 0.012 kg of carbon-12	M1 A1	[2]
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	(b) $pV = NkT$ or $pV = nRT$ substitutes temperature as 298 K <i>either</i> $1.1 \times 10^5 \times 6.5 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times 298$ <i>or</i> $1.1 \times 10^5 \times 6.5 \times 10^{-2} = n \times 8.31 \times 298$ and $n = N / 6.02 \times 10^{23}$ $N = 1.7 \times 10^{24}$	C1 C1 C1 A1	[4]
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2	(c) acceleration / force proportional to displacement from a fixed point	M1	
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